CSCI 5707 Fall 2023 – Homework 3 solutions

Answer 21.2

1. Show the view definition statements for EmployeeNames and DeptInfo.
   a.
   ```
   CREATE VIEW EmployeeNames (ename) AS
   SELECT E.ename
   FROM Employees E
   ```
   b.
   ```
   CREATE VIEW DeptInfo (dept, avgsalary) AS
   SELECT E.dept, AVG(E.salary) AS avgsalary
   FROM Employees E
   GROUP BY E.dept
   ```

2. User should be granted SELECT privilege on the view DeptInfo. Note that it is impossible to allow user to access only the average salaries of 'Toy' and 'CS' departments but not those of other departments. If we really want the average salaries of other departments to be hidden from this user, we have no choice but to create another view.

3. What privileges should you grant?
   a. To fire people, the secretary needs DELETE privileges on Employees.
   b. To check on who is an employee, the secretary needs SELECT privileges on the EmployeeNames view.
   c. To check on average department salaries, the secretary needs SELECT privileges on the DeptInfo view.

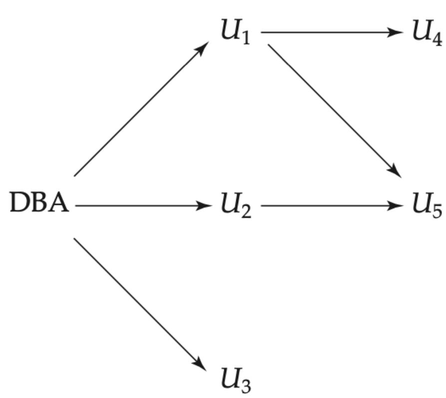
4. Given only the above privileges, it is not possible for the secretary to find out the salary of an employee using the relations alone. If the tuples are such that there is just one employee in a department and the secretary knows this information along with the name of the employee who works there, then he can possibly find out the salary. However, the relations themselves do not allow the secretary to deduce such a fact.

5. `GRANT SELECT ON EmployeeNames TO Secretary WITH GRANT OPTION`

6. `GRANT INSERT ON Employees TO Secretary`
    a. The secretary can now also insert tuples into AtoRNames, which is an updatable view created by the secretary. However, the secretary still cannot insert tuples into HowManyNames as this view is not updatable.
    b. The secretary still does not have select privileges on Employees, so they cannot see salaries for existing employees.

7. In most cases, Todd's privileges (granted by the secretary) will be revoked. Todd's privileges granted by other users will be retained. However, some databases will raise an error if a user is still in the authorization graph.

    Remarks. An authorization graph refers to a directed graph between users where the root is the database administrator(DBA) and a directed edge representing the privilege granted by the source node to the destination node.



Authorization-grant graph ($U_1, U_2, \ldots, U_5$ are users and DBA refers to the database administrator).

A user has an authorization (of some actions) if and only if there is a path from the root of the authorization graph.

8. Changing average salary for a department cannot be implemented since one doesn't know which salaries to change.

9. Statements:

a. `GRANT SELECT, UPDATE, INSERT ON Employees TO Joe WITH GRANT OPTION`
b. `GRANT SELECT, UPDATE, INSERT ON EmployeeNames TO Joe WITH GRANT OPTION`
c. No, Joe is not automatically given read access to the DeptInfo view. However, he has the equivalent access to the data since he has SELECT privileges on Employees. He could create an identical view as he has the privilege on all underlying tables.

10. There is no way to do this in SQL: even though you granted privilege to Joe and Joe granted privileges to Mike, you cannot revoke Mike's access. Either Joe has to do it, or you have to revoke Joe's access (which in turn revoke Mike's) and then grant Joe privilege again.

11. Since you don't own AllNames, you can only prevent Mike and Susan from accessing it by revoking Joe's right to read EmployeeNames:
   a. `REVOKE SELECT ON EmployeeNames FROM Joe`
   b. The view AllNames is dropped as a consequence. Joe can still modify EmployeeNames but cannot read records. Joe also has his original access to the Employees relation, which means he can build views identical to EmployeeNames and AllNames.

Answer 25.2

1. Show the result of pivoting the relation on *pid* and *timeid*.

Pid / Timeid	11	12	13	**Total**
1	60	56	28	**144**
2	30	65	50	**145**
3	25	70	15	**110**
Total	**115**	**191**	**93**	**399**

2. Write a collection of SQL queries to obtain the same result as in the previous part.

```
SELECT    S.pid, S.timeid, SUM(S.sales)
```

```
FROM      Sales S
GROUP BY    S.pid, S.timeid

SELECT    S.pid, SUM(S.sales)
FROM      Sales S
GROUP BY  S.pid

SELECT    S.timeid, SUM(S.sales)
FROM      Sales S
GROUP BY     S.timeid

SELECT   SUM(S.sales)
FROM     Sales S
```

3. Show the result of pivoting the relation on *pid* and *locid*.

Locid \ Pid	11	12	13	**Total**
1	48	100	28	**176**
2	67	91	65	**223**
Total	**115**	**191**	**93**	**399**

Answer 25.4

1. What is the difference between the WINDOW clause and the GROUP BY clause?

   - A group by clause is used to group the results of a query based of on the specified predicate, such that SQL only aggregates the result and automatically make sure that only distinct values are allowed. A window function is a function whose result for a given row is derived from the window frame of that row, which is found using the PARTITION clause. The WINDOW clause doesn't actually apply the DISTINCT clause on the result provided.
   - GROUP BY queries make use of the HASH GROUP BY operation, while Window clauses instead utilize a WINDOW SORT operation.
   - WINDOW clauses are faster than the GROUP BY clause. Also, unlike GROUP BY, the WINDOW clause has no grouping effect.
   - GROUP BY clauses work on the data directly fetched from the table, while WINDOW clauses work on the resultant data fetched by the query.

2. Give an example query that cannot be expressed in SQL without the WINDOW clause but that can be expressed with the WINDOW clause.

   Rank all products by total sales over the last year, and, for each product, print the difference in total sales relative to the product ranked behind it.

3. What is the frame of a window in SQL: 1999

   The boundaries of the window associated with each row in terms of the ordering of rows within partitions.

4. Consider the following simple GROUP BY query.

```
SELECT T.year, SUM(S.sales)
FROM Sales S, Times T
WHERE S.timeid='T.timeid GROUP BY T.year
```

   Can you write this query in SQL:1999 without using a GROUP BY clause?

```
SELECT T.year, SUM(S.sales) OVER W AS movsum
FROM Sales S, Times T
WHERE S.timeid = T.timeid
WINDOW W AS( PARTITION BY T.year ORDER BY T.year)
```

Answer 25.10

1. Following factors need to be considered to decide whether to materialize a view
   - Whether it would be possible to exploit the materialized view to answer a query on a view.
   - What indexes should be built on the materialized view.
   - How the materialized views may be synchronized with changes to the underlying table.
   - Disk space
   - Cost of refreshing
2.
   a. The auxiliary information maintained by the incremental view algorithm is number of times a row is derived in the view. This will in turn determine whether a row should be a part of the materialized view or not. Moreover, if the count reduces to zero, it means that the particular row must be deleted from the view. The main idea behind incremental maintenance algorithms is to efficiently compute changes to rows of the view, either new rows or changes to the count associated with a row.
   b. **Pros**: Data is indexed, making queries faster and easy to retrieve.
   Cons: Maintenance, disk space usage, data is outdated, and updates are costly.
3. One reason is if you want a fast response time for the content of the view. But remember that it might become inconsistent, when the table is updated.
4.
   a. We can store the information about the information recorded since the last view refresh.

b. **Pros:** The view already has been computed using a group by clause and thus the aggregation has been done which means direct querying on this data would be much faster as compared to doing the group by computation again and again. **Cons:** Similar to 2(b).

Answer 26.2
Consider the Purchases table shown in Figure 26.1.

Transid	*CustId*	*Date*	*Item*	*Qty*
111	201	5/1/99	pen	2
111	201	5/1/99	Ink	1
111	201	5/1/99	milk	3
111	201	5/1/99	juice	6
112	105	6/3/99	pen	1
112	105	6/3/99	ink	1
112	105	6/3/99	milk	1
113	106	5/10/99	pen	1
113	106	5/10/99	milk	1
114	201	6/1/99	pen	2
114	201	6/1/99	ink	2
114	201	6/1/99	juice	4
114	201	6/1/99	water	1

Figure 26.1: The Purchases Relation

1. **Simulate the algorithm for finding frequent itemsets on the table in Figure 26.1 with minsup=90%, and then find association rules with minconf=90%.**
   First find the support of item sets with one item
   - {pen} = 100%
   - {ink} = 75%
   - {milk} = 75%
   - {juice} = 50%
   - {water} = 25%

   We can say now that only the {pen} itemset has minsup = 90% and no other one element itemset. Because there is no more one element itemset with minsup = 90% we can conclude that there does not exist, and two element or further item set with minsup = 90%

   Association rules with mincon = 90% are nothing because there only exist one one element itemset. Also, it was not mentioned in the book and thus I did not include it but, a set has association with itself because that association is trivial with con = 100% and thus will be included if considered an association.

2. **Can you modify the table so that the same frequent itemsets are obtained with minsup=90% as with minsup=70% on the table shown in Figure 26.1?**
   One example possibility is shown below where support for support for the items {ink} and {milk} is reduced to below 75%.

Transid	CustId	Date	Item	Qty
111	201	5/1/99	pen	2
111	201	5/1/99	milk	3
111	201	5/1/99	juice	6
112	105	6/3/99	pen	1
112	105	6/3/99	ink	1
113	106	5/10/99	pen	1
113	106	5/10/99	milk	1
114	201	6/1/99	pen	2
114	201	6/1/99	ink	2
114	201	6/1/99	juice	4
114	201	6/1/99	water	1

3. **Simulate the algorithm for finding frequent itemsets on the table in Figure 26.1 with minsup=10% and then find association rules with minconf=90%.**
   First find the support for one element itemset (all these itemsets have minsup = 10%)
   - {pen} = 100%
   - {ink} = 75%
   - {milk} = 75%
   - {juice} = 50%
   - {water} = 25%

   Then the support of two elements itemsets are:
   - {pen, ink} = 75%
   - {pen, milk} = 75%
   - {pen, juice} = 50%
   - {pen, water} = 25%

- {ink, milk} = 50%
- {ink, juice} = 50%
- {ink, water} = 25%
- {milk, juice} = 25%
- {milk, water} = 0%
- {juice, water} = 25%

Now all the two element itemsets have minsup = 10%. Then the support of three elements itemsets are:

- {pen, ink, milk} = 50%
- {pen, ink, juice} = 50%
- {pen, ink, water} = 25%
- {pen, milk, juice} = 25%
- {pen, juice, water} = 25%
- {ink, milk, juice} = 25%
- {ink, juice, water} = 25%

  Note: Since {milk, water} has support of 0% we will not consider its supersets since they will have 0% support as well

Now only using the three element itemsets with the minsup = 10% we can get the support for 4 element itemsets

- {pen, ink, milk, juice} = 25%
- {pen, ink, juice, water} = 25%

Since {milk, water} has 0% support, there cannot be a 5-itemset having support > 0%. So, the final answer contains the following itemsets and their support:

- {pen} = 100%
- {ink} = 75%
- {milk} = 75%
- {juice} = 50%
- {water} = 25%
- {pen, ink} = 75%
- {pen, milk} = 75%
- {pen, juice} = 50%
- {pen, water} = 25%
- {ink, milk} = 50%
- {ink, juice} = 50%
- {ink, water} = 25%
- {milk, juice} = 25%
- {juice, water} = 25%

- {pen, ink, milk} = 50%
- {pen, ink, juice} = 50%
- {pen, ink, water} = 25%
- {pen, milk, juice} = 25%
- {pen, juice, water} = 25%
- {ink, milk, juice} = 25%
- {ink, juice, water} = 25%
- {pen, ink, milk, juice} = 25%
- {pen, ink, juice, water} = 25%

Association rules with confidence >= 90%: all the rules listed have confidence = 100%

- {ink} → {pen}
- {milk} → {pen}
- {juice} → {pen}
- {juice} → {ink}
- {juice} → {pen, ink}
- {water} → {pen}
- {water} → {ink}
- {water} → {juice}
- {water} → {pen, ink}
- {water} → {pen, juice}
- {water} → {ink, juice}
- {water} → {pen, ink, juice}
- {ink, milk} → {pen}
- {ink, juice} → {pen}
- {ink, water} → {pen}
- {ink, water} → {juice}
- {juice, water} → {ink}
- {juice, water} → {pen}
- {milk, juice} → {pen}
- {milk, juice} → {ink}
- {pen, juice} → {ink}
- {pen, water} → {ink}
- {pen, water} → {juice}
- {milk, juice} → {pen, ink}
- {pen, water} → {ink, juice}
- {ink, water} → {pen, juice}
- {juice, water} → {pen, ink}

4. **Can you modify the table so that the same frequent itemsets are obtained with minsup=10 percent as with minsup=70 percent on the table shown in Figure 26.1?**
   Currently the support is defined as follows:
   - {pen} - 4/4
   - {ink} - 3/4
   - {milk} - 3/4
   - {pen, ink} - 3/4
   - {pen, milk} - 3/4
   - {ink, milk} - 2/4

   To make sure that we don't decrease the support of either of {pen, milk} or {pen, ink} because of just adding only one type of this itemset we need to add both, which would result in one example as follows:
   - {111, 201, 5/1/99, pen, 2}
   - {112, 105, 5/1/99, pen, 2}
   - {113, 106,5/10/99, pen, 1}
   - {113, 106,5/10/99, ink, 1}
   - {114, 201, 6/1/99, milk, 2}
   - {114, 201, 6/1/99, pen, 2}

Answer 26.8. Consider the SubscriberInfo Relation shown in Figure 26.17. It contains information about the marketing campaign of the DB Aficionado magazine. The first two columns show the age and salary of a potential customer and the subscription column shows whether the person subscribes to the magazine. We want to use this data to construct a decision tree that helps predict whether a person will subscribe to the Magazine.

Age	Subscription	
	Yes	No
32	0	1
35	1	0
37	0	1
39	1	0
40	0	1
43	1	0

52	1	0
55	1	0
56	1	0

1. Construct the AVC-group of the root node of the tree.

Salary	Subscription	
	Yes	No
43k	1	0
45k	0	1
50k	1	0
54k	0	1
58k	0	1
68k	1	0
70k	1	0
85k	1	0
90k	1	0

2. **Assume that the splitting predicate at the root node is age <= 50. Construct the AVC groups of the two children nodes of the root node.**
Child Node 1:

Age	Subscription	
	Yes	No
32	0	1
35	1	0
37	0	1
39	1	0
40	0	1
43	1	0

Salary	Subscription	
	Yes	No
45k	0	1
54k	0	1
58k	0	1
68k	1	0
70k	1	0
90k	1	0

Child Node 2:

Age	Subscription	
	Yes	No
52	1	0
55	1	0
56	1	0

Salary	Subscription	
	Yes	No
43k	1	0
50k	1	0
85k	1	0

Answer 26.9

(The radius here is defined as the max distance between each record and its center, not the variance definition in the book)

1. Center (15.83, 143.33), Radius ≈ 105.8465

2. (i) first 3, Center (17.67, 159), Radius≈104.5456

   (ii) last 3, Center (14,127.67), Radius≈121.5968

3. Any reasonable answer. e.g., cohesion/separation, max radius, silhouette score, purity/entropy.

Answer 26.10. Assume you are given the three sequences (1, 3, 4), (2, 3, 2), (3, 3, 7). Compute the Euclidean Norm between all pairs of sequences.

a. $||(1, 3, 4) - (2, 3, 2)|| = \sqrt{(1-2)^2 + (3-3)^2 + (4-2)^2} = \sqrt{5} = 2.236$
b. $||(1, 3, 4) - (3, 3, 7)|| = \sqrt{(1-3)^2 + (3-3)^2 + (4-7)^2} = \sqrt{13} = 3.605$
c. $||(3, 3, 7) - (2, 3, 2)|| = \sqrt{(3-2)^2 + (3-3)^2 + (7-2)^2} = \sqrt{26} = 5.099$